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INDIAN INSTITUTE OF TECHNOLOGY DELHI
ABU DHABI

Stem Cells

Nature's ultimate programmable building blocks

Lecture 22

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Potency: how many cell types can a cell produce?

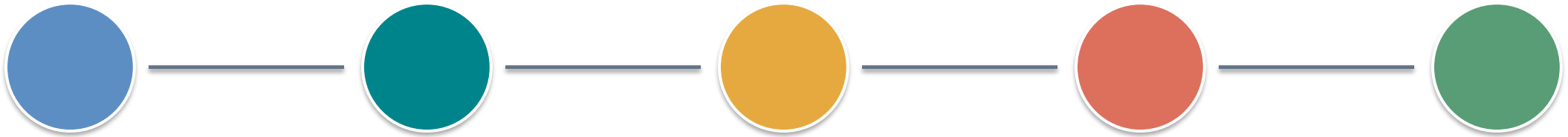
Totipotent Every cell type, including extraembryonic tissue such as placenta. Example: the zygote.

Pluripotent Can generate nearly all body cell types

Multipotent Can generate several related cell types. eg: blood stem cells make all blood cells, not neurons.

Unipotent A single cell type.

From zygote to specialized cells



Zygote
one cell

Cleavage
2–8 cells

Morula
compact cell ball

Blastocyst
inner + outer cells

Gastrulation
three germ layers

Development changes cell fate

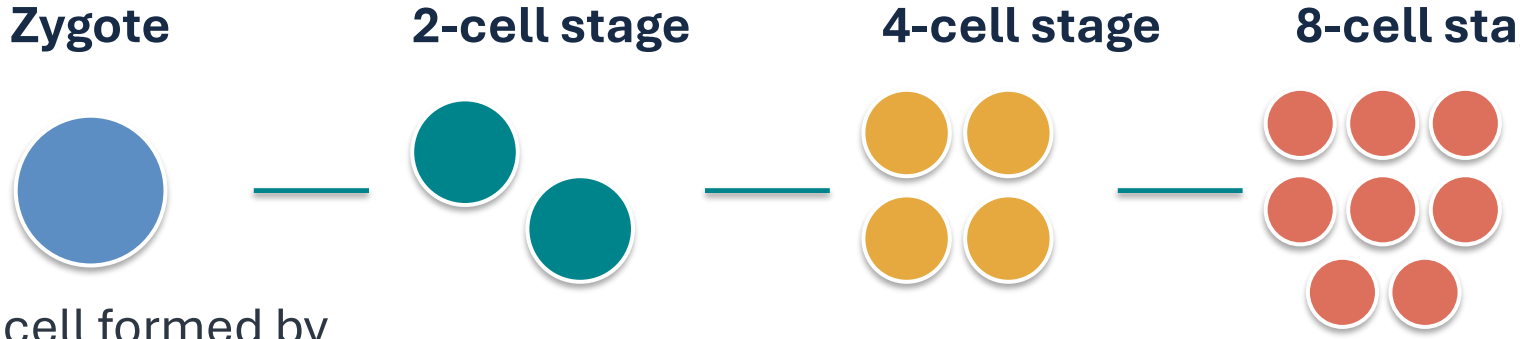
Cells divide, become organized, and gradually specialize.

Key idea

More cells does not mean more stem cells: potency and identity change over time.

Zygote and cleavage

Rapid mitotic divisions increase cell number before the embryo grows substantially



One cell formed by fertilization

Cleavage

- Repeated mitotic divisions.
- Cells are called blastomeres.
- The embryo initially grows little in overall size.

DNA replication

DNA is copied before mitosis so daughter cells receive genomes.

Cell potential

Early blastomeres have broad developmental potential, which changes with stage.

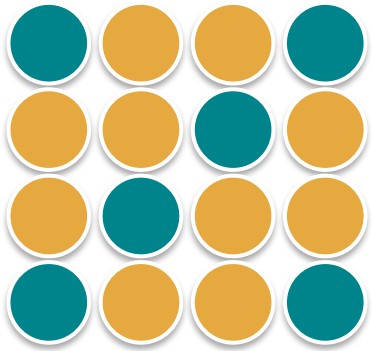
Approximate duration

Cleavage stages occur over the first few days after human fertilization.

Morula to blastocyst

Cells begin to organize into distinct populations

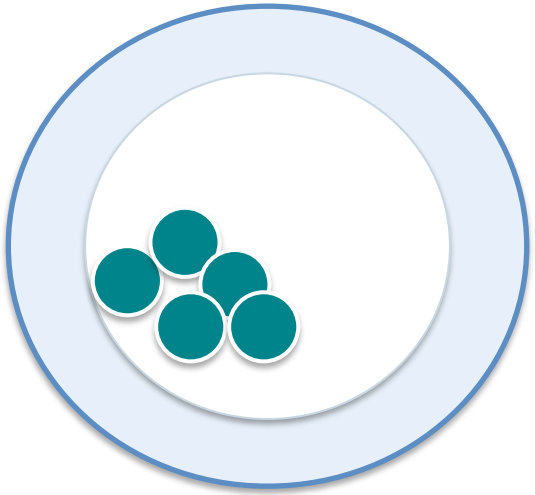
Morula



A compact ball of cells

(around day 3; approximate)

Blastocyst



Outer layer: trophectoderm

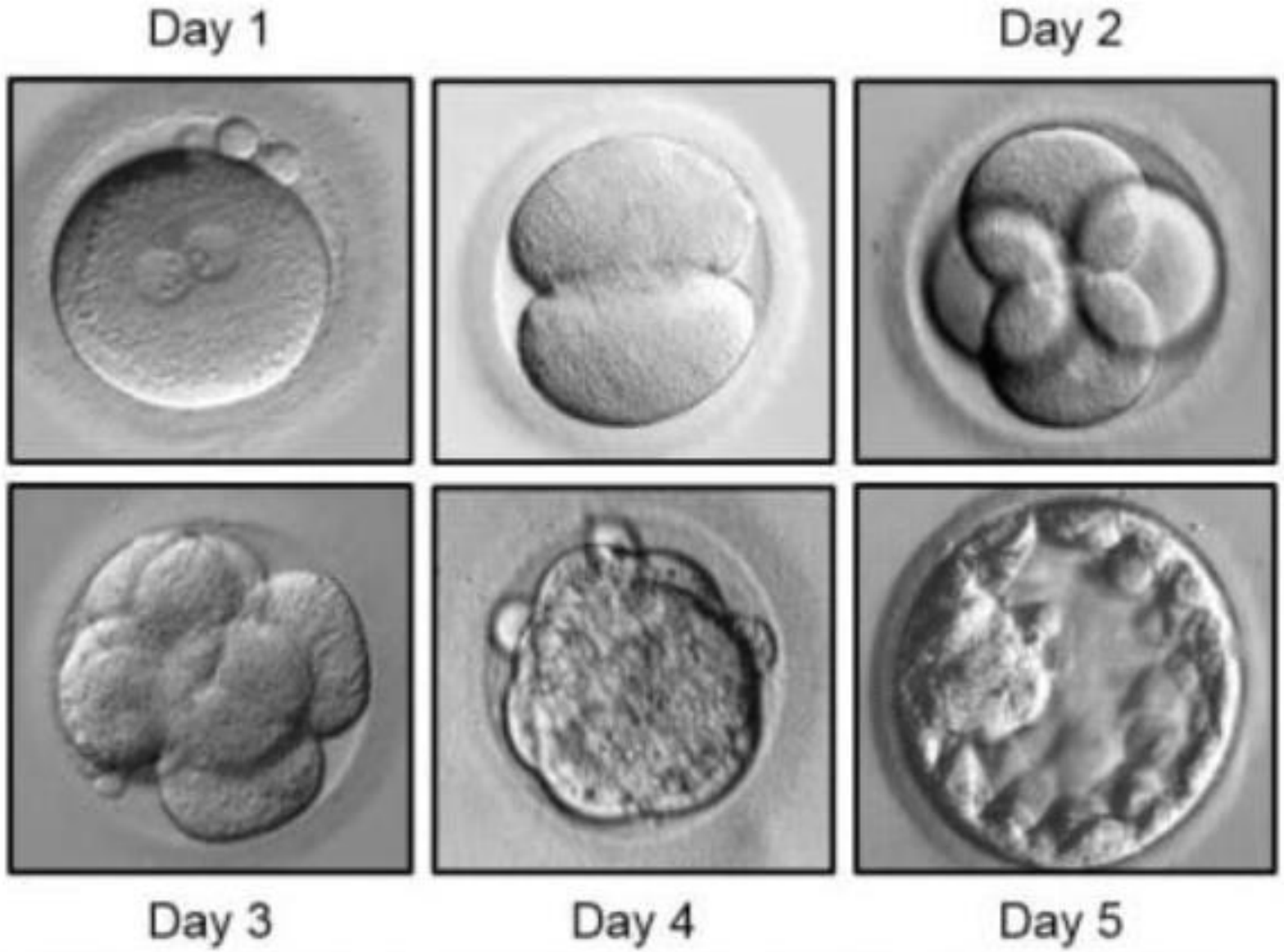
Contributes to the fetal portion of the placenta and related extraembryonic tissues.

Inner cell mass (ICM)

Gives rise to the proper embryo; embryonic stem cells derived from it are pluripotent.

The important transition

Cells are no longer all equivalent: position and signals help establish different developmental fates.



https://www.fertilityturkey.com/ivf-icsi/

Potency changes during development

Zygote	Early blastomeres	Blastocyst ICM	Tissue stem cells	Specialized cell
Totipotent	**Broad potential**	**Pluripotent**	**Multipotent**	**Differentiated**
Can contribute to embryonic and extraembryonic tissues	Potential narrows as cell-fate decisions emerge	Can form cell types of the three germ layers	Usually form a related family of cell types	Performs a defined function; usually more restricted

Potency = the *range of cell types* a cell can generate.

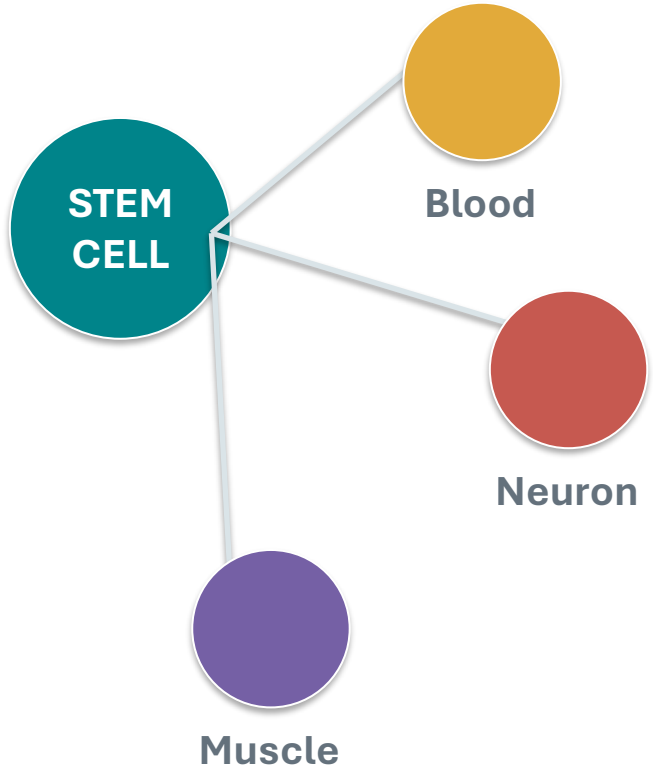
What makes a cell a stem cell?

Self-renewal

Divides over long periods without losing its potential to differentiate, and without senescence (biological ageing).

Differentiation

Gives rise to specialized cell types, such as skin, muscle or nerve cells.

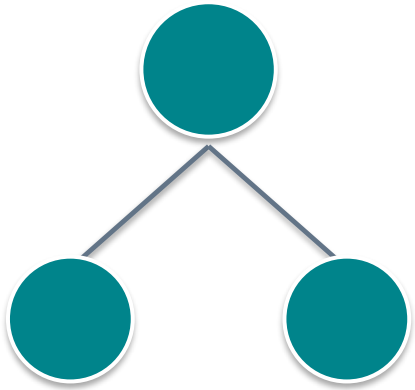


An uncommitted, blank precursor that stays unspecialized until it receives a signal. Naturally, it acts as the body's repair system.

How do stem cells divide?

Division can preserve the stem-cell pool while also supplying cells that differentiate

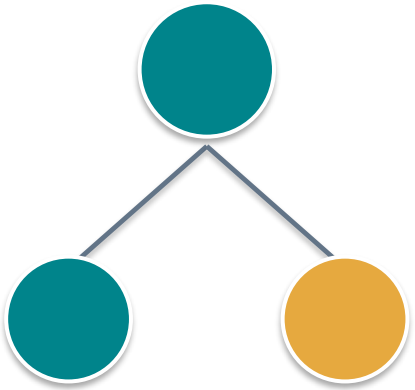
Symmetric self-renewal



Two stem cells

Expands the stem-cell pool

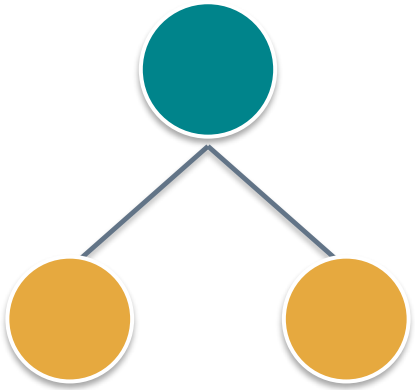
Obligate Asymmetric division



One stem cell + one progenitor

Maintains pool while supplying cells

Symmetric differentiation



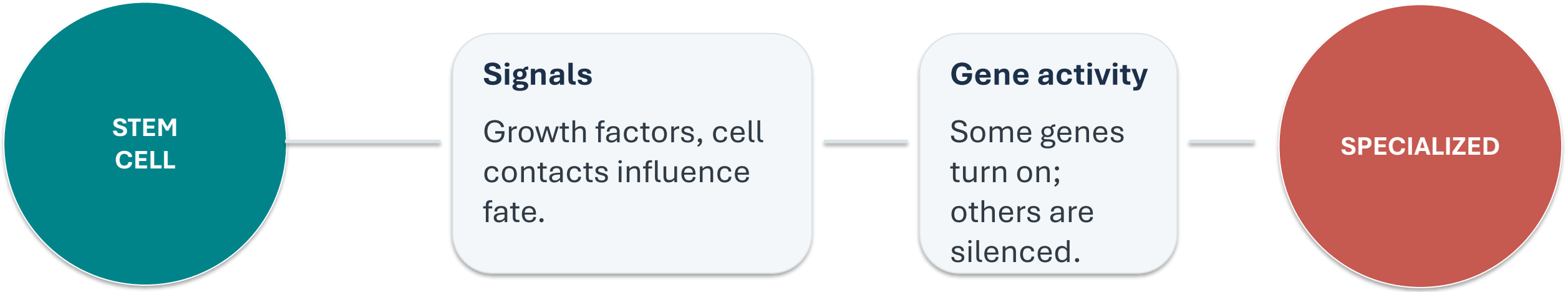
Two differentiating daughters

Can reduce the stem-cell pool

What influences the outcome?

Intrinsic cell programs and niche signals: growth factors, neighboring cells, extracellular matrix and mechanical cues.

Differentiation: from flexible to specialized



Specialized cells develop characteristic structures and functions—for example, neurons transmit signals and red blood cells carry oxygen.

Where do stem cells come from?

Embryonic (ES)

Origin
Inner cell mass of the blastocyst

Potency
Pluripotent

Trade-off
Most flexible, but ethically contested

Adult (somatic)

Origin
Tissues such as bone marrow, brain, gut and skin

Potency
Multipotent

Trade-off
Safer and less contested, but limited

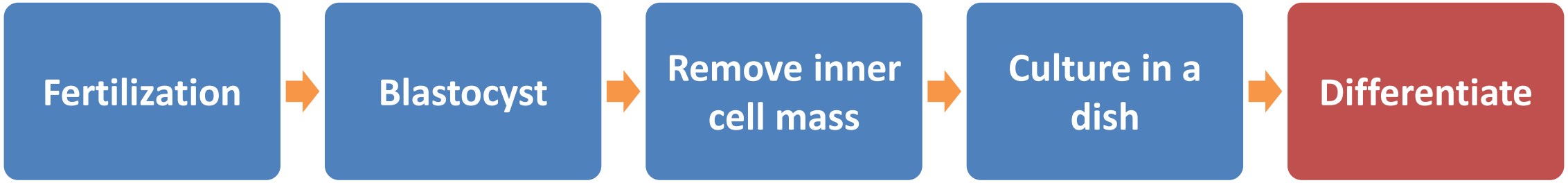
Induced pluripotent

Origin
Adult cells reprogrammed by added genes

Potency
Pluripotent

Trade-off
Patient-matched and no embryo, but safety still being tested

Embryonic stem cells: deriving a cell line



Sperm and egg join

Early embryo of 50-150 cells (about day 5)

This destroys the embryo

A cell line self-renews for long periods in vitro

Change conditions to make skin, muscle or neural cells

50-150
cells in a human blastocyst

220+
cell types in the adult body that ES cells can make

~14 µm
human ES cell diameter (mouse: ~8 µm)

Embryonic stem cells: trade-offs

Advantages

- **Flexible:** can potentially make any cell type.
- **Immortal:** one line can supply endless cells with defined traits.
- **Available:** embryos come from in vitro fertilization clinics.

Disadvantages

- **Hard to control:** difficult to differentiate uniformly into a target tissue.
- **Immunogenic:** cells from a random donor are likely to be rejected.
- **Tumorigenic:** can form tumors.
- **Ethical cost:** destroys a developing human embryo.

Adult stem cells: maintain and repair tissues

Tissue stem cells are adapted to particular biological roles

Hematopoietic

Blood-forming stem cells in bone marrow generate red cells, many immune cells and platelets.

Intestinal

Stem cells in intestinal crypts continually replace cells lining the gut.

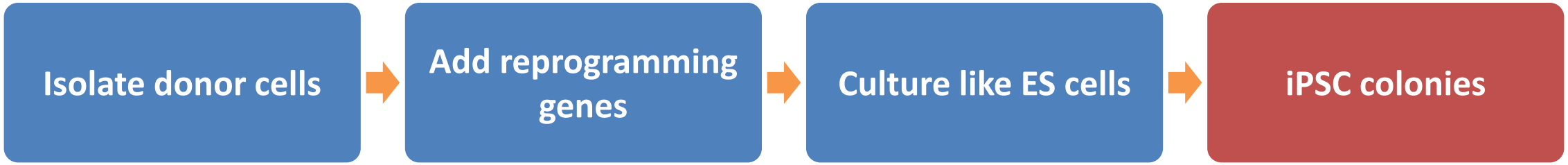
Skin / hair follicle

Stem-cell populations contribute to epidermal renewal and hair growth.

Strength: tissue maintenance. Constraint: many adult stem cells have a narrower developmental range.

Induced pluripotent stem cells (iPSCs)

Adult cells genetically reprogrammed into an embryonic-like state by forcing expression of key genes.



e.g., skin cells

iPSCs: promise and open problems

Promise

- **Disease models and drug development:** already useful tools.
- **Patient-matched tissue:** nearly identical to the donor, so rejection is probably avoided.
- **New insight:** a way to 'de-differentiate' cells and learn how to repair tissue.

Open problems

- **Viral vectors:** in animal studies they sometimes cause cancers.
- **Delivery:** non-viral methods are under investigation.
- **Equivalence:** not yet known whether iPSCs differ from ES cells in clinically significant ways.

Cell therapy: bone marrow transplant

Harvest donor marrow or stem cells → Destroy patient's leukemia cells and marrow → Inject donor stem cells → Healthy blood system regenerates

